

Big Data Analytics Viva Revision Sheet

Topic / Abbreviation	Full Form	Key Points
BDA	Big Data Analytics	Process of analyzing huge datasets for insights
5 Vs of Big Data	-	Volume, Velocity, Variety, Veracity, Value
Volume	-	Huge amount of data
Velocity	-	Speed of data generation/processing
Variety	-	Multiple formats of data
Veracity	-	Data quality and trustworthiness
Value	-	Useful insights from data
Structured Data	-	Fixed format data (tables)
Semi-Structured Data	-	JSON/XML type organized data
Unstructured Data	-	Images, videos, text, audio
Hadoop	-	Open-source distributed framework
HDFS	Hadoop Distributed File System	Distributed storage system
NameNode	-	Master node storing metadata
DataNode	-	Stores actual file blocks
YARN	Yet Another Resource Negotiator	Resource management layer
MapReduce	-	Distributed parallel processing model
Mapper	-	Converts input into key-value pairs
Reducer	-	Aggregates intermediate outputs
Combiner	-	Local reducer optimization
Shuffle & Sort	-	Groups same keys together
Replication Factor	-	Number of block copies (default = 3)
Fault Tolerance	-	System survives failures
Data Locality	-	Processing happens near data
NoSQL	Not Only SQL	Flexible schema database system
MongoDB	-	Document-oriented NoSQL database
BSON	Binary JSON	Internal MongoDB storage format
Collection	-	Equivalent to SQL table
Document	-	Equivalent to SQL row
Sharding	-	Horizontal partitioning of data
Hive	Apache Hive	Data warehouse on Hadoop
HiveQL	Hive Query Language	SQL-like language for Hive
HBase	Hadoop Database	Column-family NoSQL DB
Pig	Apache Pig	High-level data flow platform
Pig Latin	-	Language used in Pig
RDD	Resilient Distributed Dataset	Core Spark distributed dataset
Spark	Apache Spark	Fast in-memory processing engine

Topic / Abbreviation	Full Form	Key Points
DAG	Directed Acyclic Graph	Spark execution workflow
EDA	Exploratory Data Analysis	Understanding and analyzing data
EDA Steps	-	Collect → Clean → Transform → Visualize → Analyze
ML	Machine Learning	Systems learning from data
AI	Artificial Intelligence	Machines simulating intelligence
Regression	-	Predicting numerical values
Linear Regression	-	Straight-line prediction model
ANOVA	Analysis of Variance	Used to compare means of multiple groups
Histogram	-	Graph showing data distribution/frequency
Scatter Plot	-	Graph showing relationship between two variables
Box Plot	-	Graph used to detect outliers and spread
SQL	Structured Query Language	Language used for relational databases
RDBMS	Relational Database Management System	Table-based database management system
ETL	Extract Transform Load	Process of extracting, transforming, and loading data
IoT	Internet of Things	Network of smart connected devices
API	Application Programming Interface	Allows communication between software systems
CSV	Comma Separated Values	Simple tabular text file format
XML	Extensible Markup Language	Semi-structured markup-based data format
JSON	JavaScript Object Notation	Lightweight data exchange format
OLAP	Online Analytical Processing	Multidimensional analytical processing system
DataOps	Data Operations	Applying DevOps concepts to data pipelines
Streaming Analytics	-	Real-time continuous data processing and analysis
Horizontal Scaling	-	Increasing capacity by adding more machines
Vertical Scaling	-	Increasing capacity by upgrading existing machine resources

Important Comparisons

Comparison	Key Difference
SQL vs NoSQL	Fixed schema vs flexible schema
Hadoop vs Spark	Disk-based vs in-memory
Hive vs HBase	Batch analytics vs real-time
Pig vs Hive	Procedural vs declarative
Structured vs Unstructured Data	Fixed format vs no fixed format

Most Asked Viva Questions

Question	Short Answer
Explain HDFS architecture	NameNode + DataNodes + replication

Question	Short Answer
Explain MapReduce flow	Map → Shuffle → Reduce
Why Spark faster than Hadoop?	In-memory computation
What is fault tolerance?	System survives failures
What is sharding?	Horizontal partitioning
What is EDA?	Data understanding before modeling
What is RDD?	Spark immutable distributed dataset
Difference between Hive & HBase	Batch vs real-time
Why NoSQL became popular?	Scalability + flexible schema
Why replication important?	Prevents data loss